

Лабораторная работа №5

Дисциплина: Основы информационной безопасности

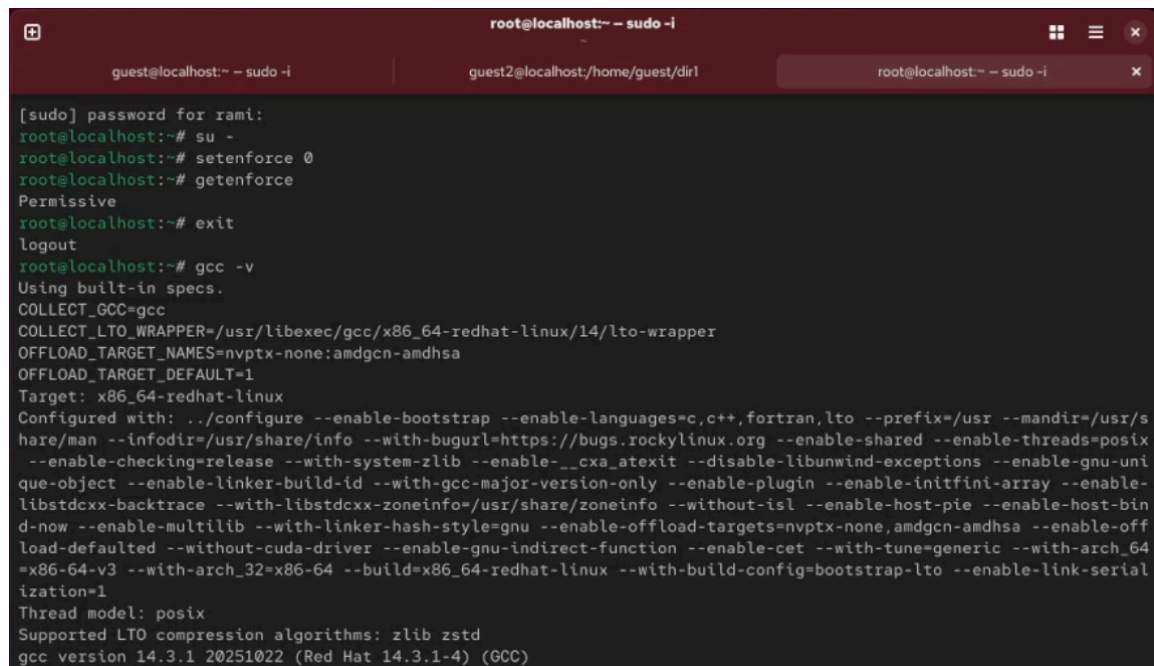
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Цель работы

Изучение механизмов изменения идентификаторов приложения SetUID и Sticky-битов. Получение практических навыков работы в консоли с дополнительными атрибутами. Рассмотрение работы механизма смены идентификатора процессов пользователей, а также влияние бита Sticky на запись и удаление файлов.

Выполнение лабораторной работы

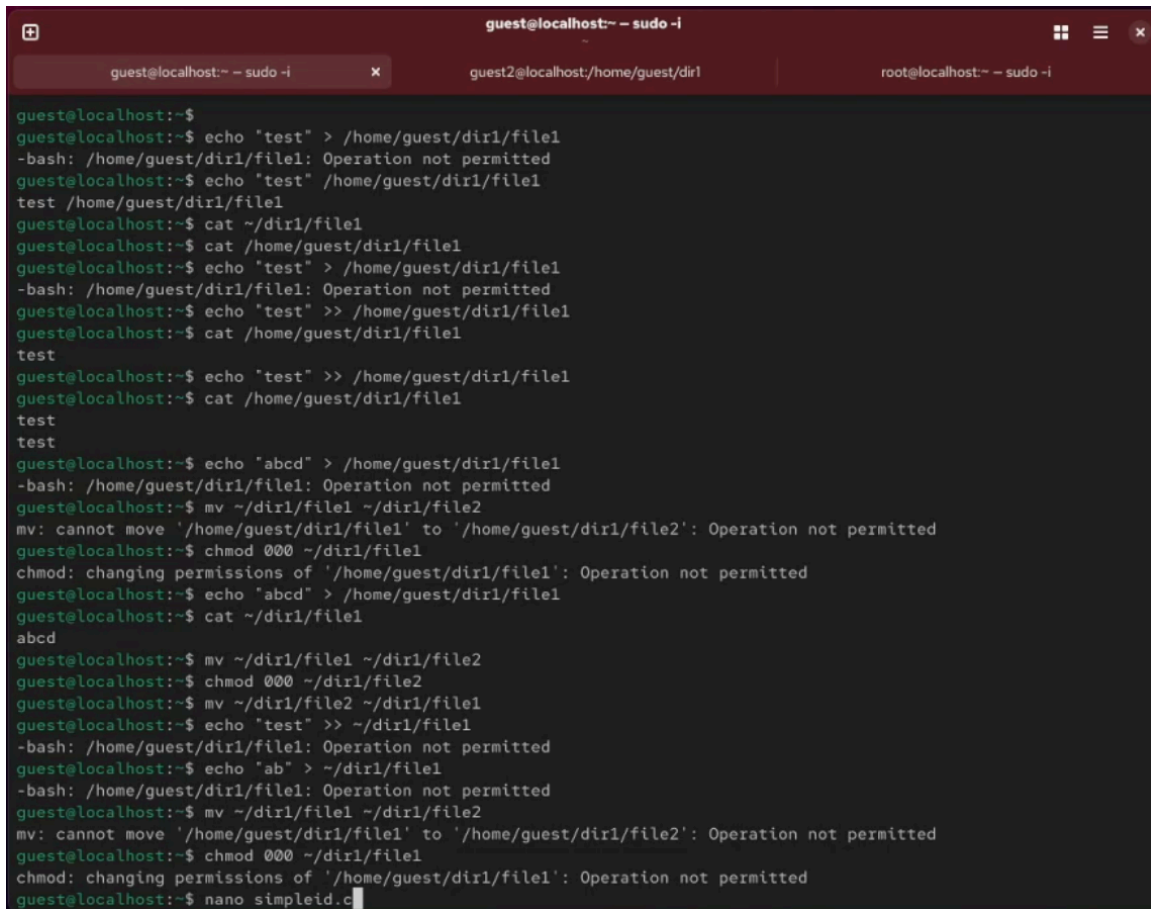
До начала работы проверил, что имеется средства разработки:



```
[sudo] password for rami:
root@localhost:~# su -
root@localhost:~# setenforce 0
root@localhost:~# getenforce
Permissive
root@localhost:~# exit
logout
root@localhost:~# gcc -v
Using built-in specs.
COLLECT_GCC=gcc
COLLECT_LTO_WRAPPER=/usr/libexec/gcc/x86_64-redhat-linux/14/lto-wrapper
OFFLOAD_TARGET_NAMES=nvptx-none:amdgc-n-amdhsa
OFFLOAD_TARGET_DEFAULT=1
Target: x86_64-redhat-linux
Configured with: ../configure --enable-bootstrap --enable-languages=c,c++,fortran,lto --prefix=/usr --mandir=/usr/s
hare/man --infodir=/usr/share/info --with-bugurl=https://bugs.rockylinux.org --enable-shared --enable-threads=posix
--enable-checking=release --with-system-zlib --enable-__cxa_atexit --disable-libunwind-exceptions --enable-gnu-uni
que-object --enable-linker-build-id --with-gcc-major-version-only --enable-plugin --enable-initfini-array --enable-
libstdc++-backtrace --with-libstdc++-zoneinfo=/usr/share/zoneinfo --without-isl --enable-host-pie --enable-host-bin
d-now --enable-multilib --with-linker-hash-style=gnu --enable-offload-targets=nvptx-none,amdgc-n-amdhsa --enable-off
load-defaulted --without-cuda-driver --enable-gnu-indirect-function --enable-cet --with-tune=generic --with-arch_64
=x86-64-v3 --with-arch_32=x86-64 --build=x86_64-redhat-linux --with-build-config=bootstrap-lto --enable-link-seria
lization=1
Thread model: posix
Supported LTO compression algorithms: zlib zstd
gcc version 14.3.1 20251022 (Red Hat 14.3.1-4) (GCC)
```

Рисунок 1. Проверка рабочее пространство

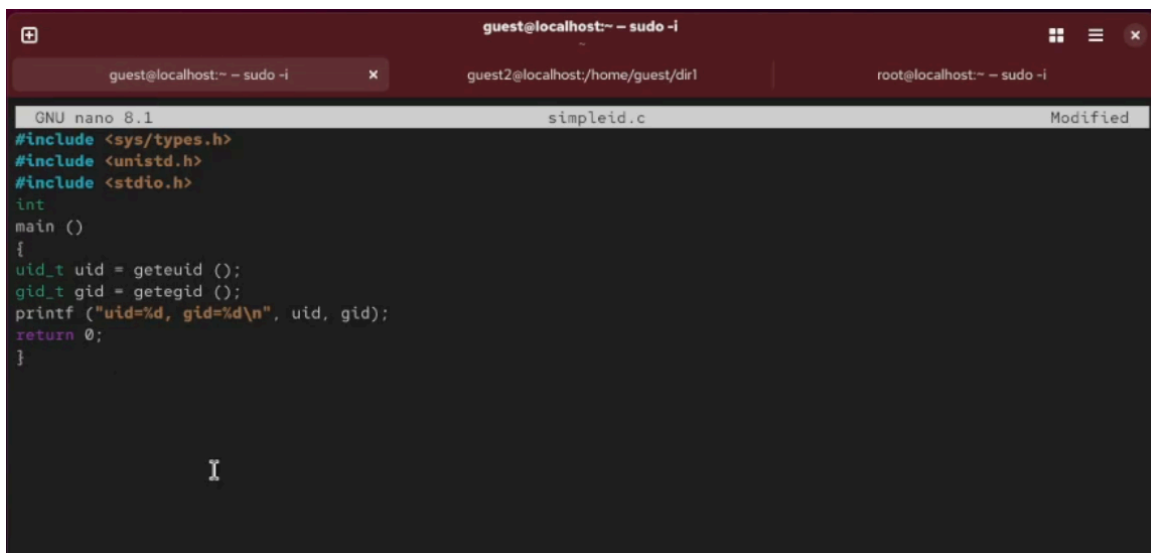
Вошел в систему от имени пользователя guest и создал программу simpleid.c:



```
guest@localhost:~ - sudo -i
guest2@localhost:/home/guest/dir1
root@localhost:~ - sudo -i

guest@localhost:~$
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" /home/guest/dir1/file1
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
guest@localhost:~$ cat /home/guest/dir1/file1
test
test
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file1
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
abcd
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
guest@localhost:~$ chmod 000 ~/dir1/file2
guest@localhost:~$ mv ~/dir1/file2 ~/dir1/file1
guest@localhost:~$ echo "test" >> ~/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "ab" > ~/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file1
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$ nano simpleid.c
```

Рисунок 2. Создание simpleid.c



```
GNU nano 8.1 simpleid.c Modified
#include <sys/types.h>
#include <unistd.h>
#include <stdio.h>
int
main ()
{
uid_t uid = geteuid ();
gid_t gid = getegid ();
printf ("uid=%d, gid=%d\n", uid, gid);
return 0;
}
```

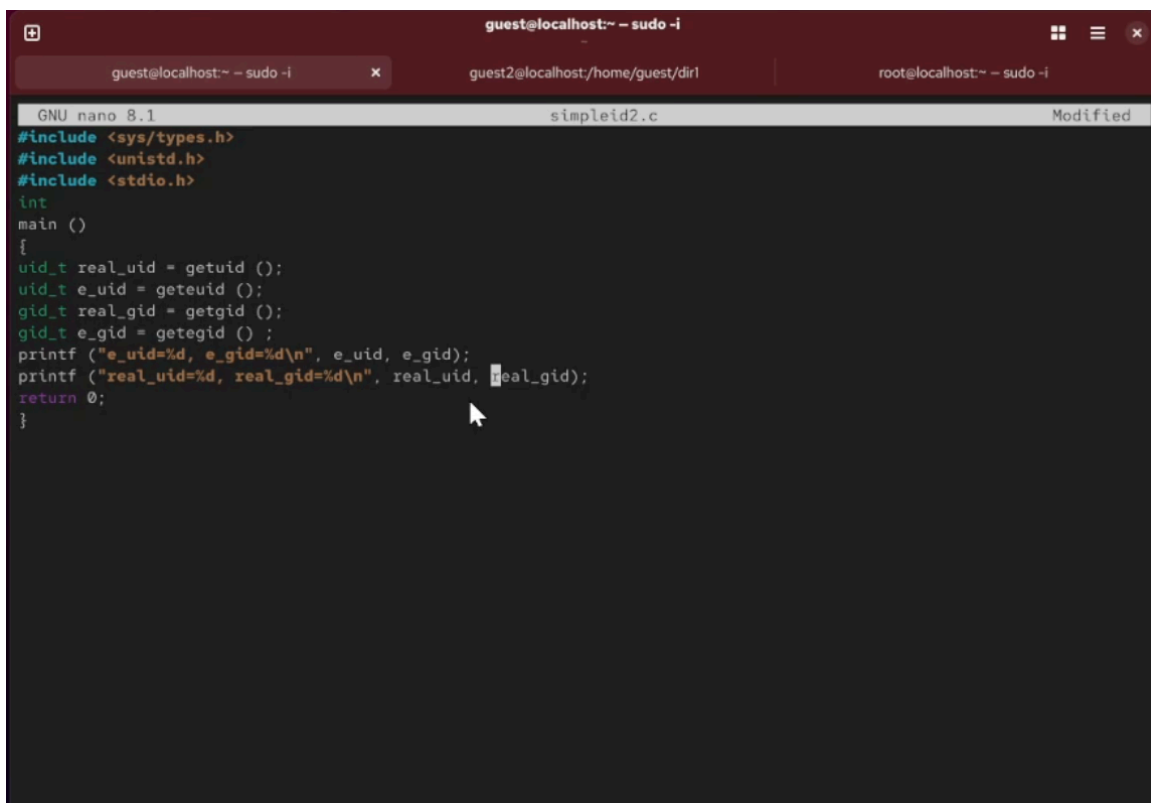
Рисунок 3. код программы 1

Скомпилировал программу и запускаю ее. Она выводит идентификатор пользователя и группы:

```
test /home/guest/dir1/file1
guest@localhost:~$ cat ~/dir1/file1
test
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "test" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
test
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "test" >> /home/guest/dir1/file1
test
guest@localhost:~$ cat /home/guest/dir1/file1
test
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file1
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$ echo "abcd" > /home/guest/dir1/file1
abcd
guest@localhost:~$ cat ~/dir1/file1
abcd
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file2
chmod: changing permissions of '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ mv ~/dir1/file2 ~/dir1/file1
mv: cannot move '/home/guest/dir1/file2' to '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$ echo "test" >> ~/dir1/file1
-bash: ~/dir1/file1: Operation not permitted
guest@localhost:~$ echo "ab" > ~/dir1/file1
-bash: ~/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file1
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$ nano simpleid.c
guest@localhost:~$ gcc simpleid.c -o simpleid
guest@localhost:~$ ./simpleid
uid=1001, gid=1001
guest@localhost:~$
```

Рисунок 4. Вывод simpleid.c

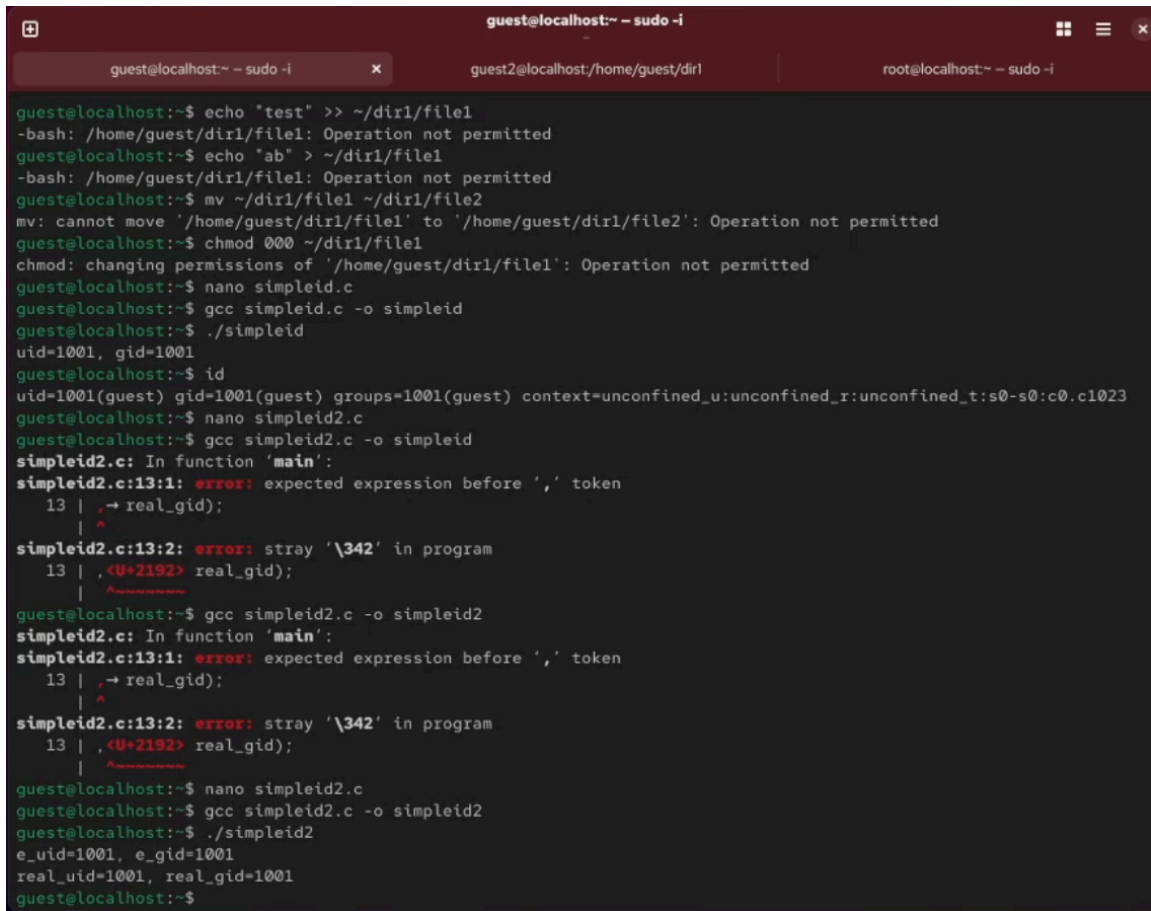
Создал файл simpleid2.c добавив вывод действительных идентификаторов.



```
GNU nano 8.1 simpleid2.c Modified
#include <sys/types.h>
#include <unistd.h>
#include <stdio.h>
int
main ()
{
uid_t real_uid = getuid ();
uid_t e_uid = geteuid ();
gid_t real_gid = getgid ();
gid_t e_gid = getegid ();
printf ("e_uid=%d, e_gid=%d\n", e_uid, e_gid);
printf ("real_uid=%d, real_gid=%d\n", real_uid, real_gid);
return 0;
}
```

Рисунок 5. simpleid2.c

Скомпилировал программу и запускаю ее.



```
guest@localhost:~$ echo "test" >> ~/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ echo "ab" > ~/dir1/file1
-bash: /home/guest/dir1/file1: Operation not permitted
guest@localhost:~$ mv ~/dir1/file1 ~/dir1/file2
mv: cannot move '/home/guest/dir1/file1' to '/home/guest/dir1/file2': Operation not permitted
guest@localhost:~$ chmod 000 ~/dir1/file1
chmod: changing permissions of '/home/guest/dir1/file1': Operation not permitted
guest@localhost:~$ nano simpleid.c
guest@localhost:~$ gcc simpleid.c -o simpleid
guest@localhost:~$ ./simpleid
uid=1001, gid=1001
guest@localhost:~$ id
uid=1001(guest) gid=1001(guest) groups=1001(guest) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
guest@localhost:~$ nano simpleid2.c
guest@localhost:~$ gcc simpleid2.c -o simpleid
simpleid2.c: In function 'main':
simpleid2.c:13:1: error: expected expression before ',' token
   13 | ,-> real_gid);
      | ^
simpleid2.c:13:2: error: stray '\342' in program
   13 | ,<U+2192> real_gid);
      | ^~~~~~
guest@localhost:~$ gcc simpleid2.c -o simpleid2
simpleid2.c: In function 'main':
simpleid2.c:13:1: error: expected expression before ',' token
   13 | ,-> real_gid);
      | ^
simpleid2.c:13:2: error: stray '\342' in program
   13 | ,<U+2192> real_gid);
      | ^~~~~~
guest@localhost:~$ nano simpleid2.c
guest@localhost:~$ gcc simpleid2.c -o simpleid2
guest@localhost:~$ ./simpleid2
e_uid=1001, e_gid=1001
real_uid=1001, real_gid=1001
guest@localhost:~$
```

Рисунок 6. Вывод simpleid2.c

С помощью `chown` изменяю владельца файла на суперпользователя, с помощью `chmod` изменяю права доступа

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:/home/guest/dir1
root@localhost:~ - sudo -i

root@localhost:~# setenforce 0
root@localhost:~# getenforce
Permissive
root@localhost:~# exit
logout
root@localhost:~# gcc -v
Using built-in specs.
COLLECT_GCC=gcc
COLLECT_LTO_WRAPPER=/usr/libexec/gcc/x86_64-redhat-linux/14/lto-wrapper
OFFLOAD_TARGET_NAMES=nvptx-none:amdgc-n-amdhsa
OFFLOAD_TARGET_DEFAULT=1
Target: x86_64-redhat-linux
Configured with: ../configure --enable-bootstrap --enable-languages=c,c++,fortran,lto --prefix=/usr --mandir=/usr/s
hare/man --infodir=/usr/share/info --with-bugurl=https://bugs.rockylinux.org --enable-shared --enable-threads=posix
--enable-checking=release --with-system-zlib --enable-__cxa_atexit --disable-libunwind-exceptions --enable-gnu-uni
que-object --enable-linker-build-id --with-gcc-major-version-only --enable-plugin --enable-initfini-array --enable-
libstdcxx-backtrace --with-libstdcxx-zoneinfo=/usr/share/zoneinfo --without-isl --enable-host-pie --enable-host-bin
d-now --enable-multilib --with-linker-hash-style=gnu --enable-offload-targets=nvptx-none,amdgc-n-amdhsa --enable-off
load-defaulted --without-cuda-driver --enable-gnu-indirect-function --enable-cet --with-tune=generic --with-arch_64
=x86-64-v3 --with-arch_32=x86-64 --build=x86_64-redhat-linux --with-build-config=bootstrap-lto --enable-link-serial
ization=1
Thread model: posix
Supported LTO compression algorithms: zlib zstd
gcc version 14.3.1 20251022 (Red Hat 14.3.1-4) (GCC)
root@localhost:~# gedit -v
bash: gedit: command not found...
root@localhost:~# yum install gedit
Last metadata expiration check: 1:56:30 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# dnf install gedit
Last metadata expiration check: 1:57:57 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# chown root:guest /home/guest/simpleid2
root@localhost:~# chmod u+s /home/guest/simpleid2
root@localhost:~#
```

Рисунок 7. Изменение права доступа на simpleid2.c

Сравнение вывода программы и команды id, наша программа вывела только ограниченное количество информации

```
guest@localhost:~ - sudo -i
guest@localhost:~$ nano simpleid.c
guest@localhost:~$ gcc simpleid.c -o simpleid
guest@localhost:~$ ./simpleid
uid=1001, gid=1001
guest@localhost:~$ id
uid=1001(guest) gid=1001(guest) groups=1001(guest) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
guest@localhost:~$ nano simpleid2.c
guest@localhost:~$ gcc simpleid2.c -o simpleid
simpleid2.c: In function 'main':
simpleid2.c:13:1: error: expected expression before ',' token
13 | ,→ real_gid);
    | ^
simpleid2.c:13:2: error: stray '\342' in program
13 | , <U+2192> real_gid);
    | ^~~~~~
guest@localhost:~$ gcc simpleid2.c -o simpleid2
simpleid2.c: In function 'main':
simpleid2.c:13:1: error: expected expression before ',' token
13 | ,→ real_gid);
    | ^
simpleid2.c:13:2: error: stray '\342' in program
13 | , <U+2192> real_gid);
    | ^~~~~~
guest@localhost:~$ nano simpleid2.c
guest@localhost:~$ gcc simpleid2.c -o simpleid2
guest@localhost:~$ ./simpleid2
e_uid=1001, e_gid=1001
real_uid=1001, real_gid=1001
guest@localhost:~$ ls -l simpleid2
-rwsr-xr-x. 1 root guest 16920 Aug 29 20:52 simpleid2
guest@localhost:~$ ./simpleid2
e_uid=0, e_gid=1001
real_uid=1001, real_gid=1001
guest@localhost:~$ id
uid=1001(guest) gid=1001(guest) groups=1001(guest) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
guest@localhost:~$
```

Рисунок 8. сравнение simpleid2 на id

Создал еще одну программу readfile.c

Рисунок 9. Программа readfile.c

Снова от имени суперпользователя изменил владельца файла readfile. Далее изменил права доступа так, чтобы пользователь guest не смог прочесть содержимое файла

```
Target: x86_64-redhat-linux
Configured with: ../configure --enable-bootstrap --enable-languages=c,c++,fortran,lto --prefix=/usr --mandir=/usr/s
hare/man --infodir=/usr/share/info --with-bugurl=https://bugs.rockylinux.org --enable-shared --enable-threads=posix
--enable-checking=release --with-system-zlib --enable-__cxa_atexit --disable-libunwind-exceptions --enable-gnu-uni
que-object --enable-linker-build-id --with-gcc-major-version-only --enable-plugin --enable-initfini-array --enable-
libstdcxx-backtrace --with-libstdcxx-zoneinfo=/usr/share/zoneinfo --without-isl --enable-host-pie --enable-host-bin
d-now --enable-multilib --with-linker-hash-style=gnu --enable-offload-targets=nvptx-none,amdgc-n-amdhsa --enable-off
load-defaulted --without-cuda-driver --enable-gnu-indirect-function --enable-cet --with-tune=generic --with-arch_64
=x86-64-v3 --with-arch_32=x86-64 --build=x86_64-redhat-linux --with-build-config=bootstrap-lto --enable-link-serial
ization=1
Thread model: posix
Supported LTO compression algorithms: zlib zstd
gcc version 14.3.1 20251022 (Red Hat 14.3.1-4) (GCC)
root@localhost:~# gedit -v
bash: gedit: command not found...
root@localhost:~# yum install gedit
Last metadata expiration check: 1:56:30 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# dnf install gedit
Last metadata expiration check: 1:57:57 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# chown root:guest /home/guest/simpleid2
root@localhost:~# chmod u+s /home/guest/simpleid2
root@localhost:~# exit
logout
rami@localhost:~$ ls -l simpleid2
ls: cannot access 'simpleid2': No such file or directory
rami@localhost:~$ sudo -i
[sudo] password for rami:
root@localhost:~# chmod g+s /home/guest/simpleid2
root@localhost:~# chown root:guest /home/guest/simpleid2
root@localhost:~# chmod g+s /home/guest/simpleid2
root@localhost:~# chown root:root /home/guest/readfile.c
root@localhost:~# chmod 600 /home/guest/readfile.c
root@localhost:~#
```

Рисунок 10. Смена владельца файла и прав доступа

Проверка прочесть файл от имени пользователя guest. Прочесть файл не удастся. Попытка прочесть тот же файл с помощью программы readfile, в ответ получаем «отказано в доступе»


```

guest@localhost:~ - sudo -i
size_t bytes_read;
int i;
int fd = open (argv[1], O_RDONLY);
do
{
bytes_read = read (fd, buffer, sizeof (buffer));
for (i =0; i < bytes_read; ++i) printf("%c", buffer[i]);
}
while (bytes_read == sizeof (buffer));
close (fd);
return 0;
}
guest@localhost:~$ ./readfile /etc/shadow
#####6VgPPP####LaXW^ P##Qy###eLeX###L#Z#
P#####
d6I###Ve###EYpeP####eP####8TP_P##kP###fP####P####P####P####P####P####P####P####P####P####P####P####P####PgP####P####
6!!o!3pe!!d@a8
|
9P##sP##IP####S
d62[[]/i/x86_64./readfile/etc/shadowSHELL=/bin/bashHISTCONTROL=ignoredupsHISTSIZ=100HOSTNAME=lo
calhostGPG_TTY=/dev/pts/0PWD=/home/guestLOGNAME=guestHOME=/home/guestLANG=en_US.UTF-8LS_COLORS=rs=0:di=01:34:ln=01:
36:mh=00:pi=40:33:so=01:35:do=01:35:bd=40:33:01:cd=40:33:01:or=40:31:01:mi=01:37:41:su=37:41:sg=30:43:ca=00:tw=30:4
2:ow=34:42:st=37:44:ex=01:32:*.*7z=01:31:*.*ace=01:31:*.*alz=01:31:*.*apk=01:31:*.*arc=01:31:*.*arj=01:31:*.*bz=01:31:*.*bz
2=01:31:*.*cab=01:31:*.*cpio=01:31:*.*crate=01:31:*.*deb=01:31:*.*drpm=01:31:*.*dwm=01:31:*.*dz=01:31:*.*ear=01:31:*.*egg=01
:31:*.*esd=01:31:*.*gz=01:31:*.*jar=01:31:*.*lha=01:31:*.*lrz=01:31:*.*lz=01:31:*.*lz4=01:31:*.*lzh=01:31:*.*lzma=01:31:*.*lz
o=01:31:*.*pyz=01:31:*.*rar=01:31:*.*rpm=01:31:*.*rz=01:31:*.*sar=01:31:*.*swm=01:31:*.*t7z=01:31:*.*tar=01:31:*.*taz=01:31:
*.tbz=01:31:*.*tbz2=01:31:*.*tgz=01:31:*.*tlz=01:31:*.*txz=01:31:*.*tz=01:31:*.*tzo=01:31:*.*tztst=01:31:*.*udeb=01:31:*.*war
=01:31:*.*whl=01:31:*.*wim=01:31:*.*xz=01:31:*.*z=01:31:*.*zip=01:31:*.*zoo=01:31:*.*zst=01:31:*.*avif=01:35:*.*jpg=01:35:*.*
jpeg=01:35:*.*mjpg=01:35:*.*mjpeg=01:35:*.*gif=01:35:*.*bmp=01:35:*.*pbm=01:35:*.*pgm=01:35:*.*ppm=01:35:*.*tga=01:35:*.*xbm
=01:35:*.*xpm=01:35:*.*tif=01:35:*.*tiff=01:35:*.*png=01:35:*.*svg=01:35:*.*svgt=01:35:*.*mng=01:35:*.*pcx=01:35:*.*mov=01:3
5:*.*mpg=01:35:*.*mpeg=01:35:*.*m2v=01:35:*.*mkv=01:35:*.*webm=01:35:*.*webp=01:35:*.*ogm=01:35:*.*mp4=01:35:*.*m4v=01:35:*.*
mp4v=01:35:*.*vob=01:35:*.*qt=01:35:*.*nuv=01:35:*.*wmv=01:35:*.*asf=01:35:*.*rm=01:35:*.*rmvb=01:35:*.*flc=01:35:*.*avi=01:
35:*.*fli=01:35:*.*flv=01:35:*.*gl=01:35:*.*dl=01:35:*.*xcf=01:35:*.*xwd=01:35:Segmentation fault (core dumped)
guest@localhost:~$ ls -l / | grep tmp
drwxrwxrwt. 17 root root 4096 Aug 29 21:02 tmp
guest@localhost:~$ echo "test" > /tmp/file01.txt
guest@localhost:~$
```

Рисунок 13. Создание файла

Добавляю права на чтение и запись для других пользователей


```
guest2@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

guest2@localhost:/home/guest/dir1$ ls
file1
guest2@localhost:/home/guest/dir1$ mv /home/guest/dir1/file1 /home/guest/dir1/file2
guest2@localhost:/home/guest/dir1$ ls
file2
guest2@localhost:/home/guest/dir1$ chmod 777 /fil2
chmod: cannot access '/fil2': No such file or directory
guest2@localhost:/home/guest/dir1$ chmod 777 file2
guest2@localhost:/home/guest/dir1$ rm file2
guest2@localhost:/home/guest/dir1$ ls
guest2@localhost:/home/guest/dir1$ touch /home/guest/dir1/file1
guest2@localhost:/home/guest/dir1$ rm file1
guest2@localhost:/home/guest/dir1$ gcc -v
Using built-in specs.
COLLECT_GCC=gcc
COLLECT_LTO_WRAPPER=/usr/libexec/gcc/x86_64-redhat-linux/14/lto-wrapper
OFFLOAD_TARGET_NAMES=nvptx-none:amdgc-n-amdhsa
OFFLOAD_TARGET_DEFAULT=1
Target: x86_64-redhat-linux
Configured with: ../configure --enable-bootstrap --enable-languages=c,c++,fortran,lto --prefix=/usr --mandir=/usr/s
hare/man --infodir=/usr/share/info --with-bugurl=https://bugs.rockylinux.org --enable-shared --enable-threads=posix
--enable-checking=release --with-system-zlib --enable-__cxa_atexit --disable-libunwind-exceptions --enable-gnu-uni
que-object --enable-linker-build-id --with-gcc-major-version-only --enable-plugin --enable-initfini-array --enable-
libstdc++-backtrace --with-libstdc++-zoneinfo=/usr/share/zoneinfo --without-isl --enable-host-pie --enable-host-bin
d-now --enable-multilib --with-linker-hash-style=gnu --enable-offload-targets=nvptx-none,amdgc-n-amdhsa --enable-off
load-defaulted --without-cuda-driver --enable-gnu-indirect-function --enable-cet --with-tune=generic --with-arch_64
=x86-64-v3 --with-arch_32=x86-64 --build=x86_64-redhat-linux --with-build-config=bootstrap-lto --enable-link-seria
lization=1
Thread model: posix
Supported LTO compression algorithms: zlib zstd
gcc version 14.3.1 20251022 (Red Hat 14.3.1-4) (GCC)
guest2@localhost:/home/guest/dir1$ cd ~
guest2@localhost:~$ cat /tmp/file01.txt
test
guest2@localhost:~$ echo "test2" >> /tmp/file01.txt
bash: /tmp/file01.txt: Permission denied
guest2@localhost:~$
```

Рисунок 15. Проверка файл на чтение и запись

От имени суперпользователя снял с директории атрибут Sticky.

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~

ization=1
Thread model: posix
Supported LTO compression algorithms: zli zstd
gcc version 14.3.1 20251022 (Red Hat 14.3.1-4) (GCC)
root@localhost:~# gedit -v
bash: gedit: command not found...
root@localhost:~# yum install gedit
Last metadata expiration check: 1:56:30 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# dnf install gedit
Last metadata expiration check: 1:57:57 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# chown root:guest /home/guest/simpleid2
root@localhost:~# chmod u+s /home/guest/simpleid2
root@localhost:~# exit
logout
rami@localhost:~$ ls -l simpleid2
ls: cannot access 'simpleid2': No such file or directory
rami@localhost:~$ sudo -i
[sudo] password for rami:
root@localhost:~# chmod g+s /home/guest/simpleid2
root@localhost:~# chown root:guest /home/guest/simpleid2
root@localhost:~# chmod g+s /home/guest/simpleid2
root@localhost:~# chown root:root /home/guest/readfile.c
root@localhost:~# chmod 600 /home/guest/readfile.c
root@localhost:~# chown root:guest /home/guest/readfile.c
root@localhost:~# chmod u+s /home/guest/readfile.c
root@localhost:~# chown root:root /home/guest/readfile.c
root@localhost:~# chmod u+s /home/guest/readfile.c
root@localhost:~# chmod 4755 /home/guest/readfile.c
root@localhost:~# chmod o+rw /tmp/file01.txt
root@localhost:~# chmod -t /tmp
bash: chmod: command not found...
root@localhost:~# chmod -t /tmp
root@localhost:~#
```

Рисунок 16. Снятие атрибута Sticky

Проверил, что атрибут действительно снят и выполнен повтор предыдущих действий.

```
guest2@localhost:~  
guest@localhost:~ - sudo -i  
guest2@localhost:~  
root@localhost:~ - sudo -i  
que-object --enable-linker-build-id --with-gcc-major-version-only --enable-plugin --enable-initfini-array --enable-libstdcxx-backtrace --with-libstdcxx-zoneinfo=/usr/share/zoneinfo --without-isl --enable-host-pie --enable-host-bind-now --enable-multilib --with-linker-hash-style=gnu --enable-offload-targets=nvptx-none,amdgc-n-amdhsa --enable-offload-defaulted --without-cuda-driver --enable-gnu-indirect-function --enable-cet --with-tune=generic --with-arch_64=x86-64-v3 --with-arch_32=x86-64 --build=x86_64-redhat-linux --with-build-config=bootstrap-lto --enable-link-serialization=1  
Thread model: posix  
Supported LTO compression algorithms: zlib zstd  
gcc version 14.3.1 20251022 (Red Hat 14.3.1-4) (GCC)  
guest2@localhost:~/home/guest/dir1$ cd ~  
guest2@localhost:~$ cat /tmp/file01.txt  
test  
guest2@localhost:~$ echo "test2" >> /tmp/file01.txt  
bash: /tmp/file01.txt: Permission denied  
guest2@localhost:~$ echo "test2" > /tmp/file01.txt  
bash: /tmp/file01.txt: Permission denied  
guest2@localhost:~$ cat /tmp/file01.txt  
test  
guest2@localhost:~$ echo "test2" >> /tmp/file01.txt  
bash: /tmp/file01.txt: Permission denied  
guest2@localhost:~$ echo "test2" >> /tmp/file01.txt  
guest2@localhost:~$ cat /tmp/file01.txt  
test  
test2  
guest2@localhost:~$ echo "test3" > /tmp/file01.txt  
guest2@localhost:~$ cat /tmp/file01.txt  
test3  
guest2@localhost:~$ rm /tmp/file01.txt  
rm: cannot remove '/tmp/file01.txt': Operation not permitted  
guest2@localhost:~$ ls -l / | grep tmp  
drwxrwxrwx. 17 root root 4096 Aug 29 21:19 tmp  
guest2@localhost:~$ echo "test2" >> /tmp/file01.txt  
guest2@localhost:~$ echo "test3" > /tmp/file01.txt  
guest2@localhost:~$ cat /tmp/file01.txt  
test3  
guest2@localhost:~$ rm /tmp/file01.txt  
guest2@localhost:~$
```

Рисунок 17. Повтор предыдущих действий

Возвращение директории tmp атрибута t от имени суперпользователя

```
root@localhost:~ - sudo -i
guest@localhost:~ - sudo -i
guest2@localhost:~
root@localhost:~ - sudo -i

Thread model: posix
Supported LTO compression algorithms: zlib zstd
gcc version 14.3.1 20251022 (Red Hat 14.3.1-4) (GCC)
root@localhost:~# gedit -v
bash: gedit: command not found...
root@localhost:~# yum install gedit
Last metadata expiration check: 1:56:30 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# dnf install gedit
Last metadata expiration check: 1:57:57 ago on Sat 29 Aug 2026 06:46:01 PM CEST.
No match for argument: gedit
Error: Unable to find a match: gedit
root@localhost:~# chown root:guest /home/guest/simpleid2
root@localhost:~# chmod u+s /home/guest/simpleid2
root@localhost:~# exit
logout
rami@localhost:~$ ls -l simpleid2
ls: cannot access 'simpleid2': No such file or directory
rami@localhost:~$ sudo -i
[sudo] password for rami:
root@localhost:~# chmod g+s /home/guest/simpleid2
root@localhost:~# chown root:guest /home/guest/simpleid2
root@localhost:~# chmod g+s /home/guest/simpleid2
root@localhost:~# chown root:root /home/guest/readfile.c
root@localhost:~# chmod 600 /home/guest/readfile.c
root@localhost:~# chown root:guest /home/guest/readfile.c
root@localhost:~# chmod u+s /home/guest/readfile.c
root@localhost:~# chown root:root /home/guest/readfile.c
root@localhost:~# chmod u+s /home/guest/readfile.c
root@localhost:~# chmod 4755 /home/guest/readfile.c
root@localhost:~# chmod o+rx /tmp/file01.txt
root@localhost:~# chmot -t /tmp
bash: chmot: command not found...
root@localhost:~# chmod -t /tmp
root@localhost:~# chmod +t /tmp
root@localhost:~#
```

Рисунок 18. Изменение атрибутов

Выводы

Изучил механизм изменения идентификаторов, применил SetUID- и Sticky-биты. Получил практические навыки работы в консоли с дополнительными атрибутами. Рассмотрел работы механизма смены идентификатора процессов пользователей, а также влияние бита Sticky на запись и удаление файлов.

Список литературы

https://esystem.rudn.ru/pluginfile.php/3096713/mod_resource/content/2/005-lab_discret_sticky.pdf